ETALE SHEAVES SATISFY DESCENT FOR A SURJECTIVE FLAT LOCALLY FINITELY PRESENTED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the equalizer quasi-inverse, the reduction to f^{-1}G = F checked locally on Y, the finite cover dominating the fppf cover, the after-base-change reduction, and the final appeal to the sections case.

Editorial note. The label lemma-glue-etale-sheaf-fppf-cover names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's descent datum notation is the author's and is delivered verbatim; the four cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The equalizer description of the quasi-inverse for the integral case. Same chapter, the lemma establishing descent for a surjective integral morphism with the equalizer construction of the glued sheaf. Supplies the candidate quasi-inverse G = Equalizer(f_*F => f_{1,*}F_1) and the reduction to proving f^{-1}G = F.
- Checking the comparison after a base change. Same chapter, the lemma allowing the verification of the gluing statement to be performed after a base change by a finite morphism. Supplies the passage from Y to a finite cover Z -> Y and to Z x_Y Z.
- Domination of an fppf cover by a finite surjective morphism. More on Morphisms, the lemma producing a finite surjective morphism which Zariski locally factors through a given surjective flat locally finitely presented morphism. Supplies the cover Z -> Y and the open covering Z = union W_i with W_i -> Y factoring through X.
- Descent in the case where the morphism has a section. Same chapter, the lemma treating the gluing problem for a morphism with a section. Supplies the final step once the morphism has sections Zariski locally.

Source reference. Original reference 0GF1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23068–23068. Prerequisite pointer invoked at proof

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line(s) 23136; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GEZ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22972–22972. Prerequisite pointer invoked at proof line(s) 23114; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GEY: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22952–22952. Prerequisite pointer invoked at proof line(s) 23142; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CP0: sources/stacks-chow/tags.txt, source lines 15910–15910. Prerequisite pointer invoked at proof line(s) 23134; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes which is surjective, flat, locally of finite presentation. The functor*

$$Sh(Y_{étale}) \longrightarrow \textit{descent data for étale sheaves wrt } \{X \to Y\}$$

is an equivalence of categories.

Proof. Exactly as in the proof of Lemma 0GEZ we claim a quasi-inverse is given by the functor sending $(\mathcal{F}, \varphi)$ to

$$\mathcal{G} = \text{Equalizer}\Big(f_*\mathcal{F} \rightrightarrows f_{1,*}\mathcal{F}_1 \Big)$$

and in order to prove this it suffices to show that $f^{-1}\mathcal{G} \to \mathcal{F}$ is an isomorphism. This we may check locally, hence we may and do assume Y is affine. Then we can find a finite surjective morphism $Z \to Y$ such that there exists an open covering $Z = \bigcup W_i$ such that $W_i \to Y$ factors through X. See More on Morphisms, Lemma 0CP0. Applying Lemma 0GF1 we see that it suffices to prove the lemma after replacing Y by Z and $Z \times_Y Z$ and f by its base change. Thus we may assume f has sections Zariski locally. Of course, using that the problem is local on Y we reduce to the case where we have a section which is Lemma 0GEY. □